

I'll begin by loading the data and producing the requested analyses. Here's the approach I'll follow:

1. **Data Overview:** Brief look at structure, size, and key columns.
2. **Global Effect of Punishment:** Compare efficiency with and without punishment.
3. **Treatment Effects in Paired Configurations:** Within each unique game config, compare efficiency with/without punishment.
4. **Heterogeneity Analysis:** Analyze how effects differ by major parameters (like MPCR, player count, chat, etc.).
5. **Predictive Modeling:** Fit at least two models (linear regression, random forest), show influential features and directionality.
6. **Practical Heuristics and Quantification:** Summarize conditions where punishment helps, and expected effect sizes by parameter range.
7. **Caveats:** Note limitations.

Let's begin by loading and reviewing the dataset.